WeBe Band: Evaluation of Threshold-Based Fall Detection Framework Using Wrist-Worn Triaxial Accelerometer Signals

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***Abstract*—Falls are a major factor contributing to injury and reduced quality of life, particularly among older adults and individuals living with long-term health conditions. This study evaluates a threshold-based fall detection framework using the WeBe Band, a wrist-worn wearable device. Data were collected from eight participants at 25 Hz, each performing six types of falls and eight types of activities of daily living (ADLs). To improve robustness, data augmentation techniques including time stretching, amplitude scaling, and Gaussian noise were applied. The algorithm detects falls based on impact, motionless, and angle thresholds optimized in each leave-one-out cross-validation (LOOCV) fold and tuned using Optuna. A global threshold set was derived by averaging optimized threshold sets across folds. Using this configuration, the method achieved an overall accuracy of 79%. The proposed framework demonstrates computational efficiency suitable for real-time embedded deployment and the potential of threshold-based models for wearable fall detection applications.**

***Index Terms*—threshold-based algorithm, embedded system, optimization, medical wearables, fall detection**

I. Introduction

Motion recognition has become a core functionality of modern wearable devices, supporting multiple applications such as step counting, activity recognition, gesture control, and fall detection. In the context of an aging population, fall detection has become one of the most important applications since falls are a major factor contributing to injury and reduced quality of life, particularly among older adults and people living with long-term health conditions. One of the most serious consequences is the long-lie, in which fallers who have been lying unintentionally for a long period of time and are unable to stand up until they are noticed and assisted by others. Studies show that 20-47% of fallers are unable to get up by themselves after falling, and half of elderly people who experience long-lie die within 6 months [1], [2]. To prevent this situation, accurate, continuous, and reliable monitoring systems are crucial.

Many existing fall-detection models rely on inertial measurement units (IMUs), combining tri-axial accelerometers and gyroscopes [8], [9]. Although IMUs can capture rich motion dynamics, they increase device complexity, cost, and power consumption [6]. In contrast, accelerometer-only models are a compact and power-efficient alternative which can still capture subtle motion dynamics of falls [3], [4], [6]. However, some studies indicate that fixed thresholds tuned on simulated data fail to generalize across subjects and real-world environments [3], [5]. This highlights the necessity for subject-aware threshold refinement methods to improve robustness. These findings suggest that efficient fall detection does not necessarily require additional sensors, but rather well-refined thresholds.

Therefore, this study aims to evaluate the generalization capability of a simple threshold-based fall detection algorithm across different subjects when deployed on a wrist-worn sensor. We optimize the thresholds using Optuna under a leave-one-out cross-validation (LOOCV) framework and evaluate the global threshold set on the full dataset to assess generalization. The algorithm uses simple threshold logic, including impact threshold, post-impact duration, motionless threshold, motionless duration, and posture-changing angle threshold, which can be efficiently computed in real time on embedded hardware.

We propose a generalizable fall detection framework using 3-axis accelerometer signals on WeBe Band, a recently introduced watch-like wearable platform for continuous motion recognition [10]. At its core, WeBe Band is equipped with an ARM Cortex-M4F microcontroller (nRF52840, 64 MHz), with Flash and RAM sizes of 1 MB and 256 KB, respectively [10].

In this study, we leverage the triaxial accelerometer signals of WeBe Band to build a fall detection algorithm. To enhance robustness, data augmentation was applied to expand the available dataset. We optimize the threshold parameters within a leave-one-out cross-validation (LOOCV) framework to derive a global threshold set by averaging the tuned threshold sets from each subject, and then evaluate it using the entire dataset to assess overall performance. The results show the potential of the proposed threshold-based model, while revealing remaining challenges associated with the wrist-worn device placement.

II. Methodology

A. Data

For this study, we exclusively use the integrated triaxial accelerometer sensor in WeBe Band. Fall data were collected at a sampling rate of 25 Hz.

B. Data Collection

Time series data were collected using the WeBe Band worn on the left wrist. Eight participants were recruited for the study. Falls are conducted on an inflatable mattress. Each participant completed five recording sessions for each activity. In each session, they performed a single activity. We considered two significantly distinct activities:

- **Falls**: Each participant did the following types of fall
  1) Sitting and Fall Left
  2) Sitting and Fall Backward
  3) Standing and Fall Backward
  4) Standing and Fall Forward
  5) Sitting and Fall Right
  6) Tripped and Fall Forward
- **Activities of Daily Living (ADLs)**: Each participant did the following types of ADL
  1) Sitting in chair and Standing up
  2) Walking
  3) Walking down stairs
  4) Walking up stairs
  5) Jogging
  6) Clapping
  7) Jumping and Reaching
  8) Tie shoe laces crouch

C. Data Annotation

Each file contains a single Fall or ADL recording, and the file name consists of three parts: subjects (S), activities (A/F), Repetition (R) (e.g. S01_F01_R01, S01_A01_R01). For the second part of the file name, it can be "A" or "F", where "A" represents Activities of Daily Living (ADLs) and "F" represents Falls. We used a custom script to extract the second part of the file name, and converted it into binary labels, where 0 represents ADLs, 1 represents Falls.

D. Data Augmentation

Data augmentation was applied to increase the variety of signal dynamics and improve robustness. The following augmentation techniques were applied to each file.

- **Time Stretching**: To address the variance of motion speed, signals were stretched by $\pm 10\%$.
- **Amplitude Scaling**: Signals were scaled by $\pm 10\%$ to simulate slight variations in fall and ADL intensity between subjects.
- **Gaussian Noise**: Zero-mean Gaussian noise with a small standard deviation (set to 2% of the signal's standard deviation) was added to signals to simulate sensor imperfections and random fluctuations, improving the robustness to real-world measurement noise.

The following subjects S01-S08 represent the original ones, while S09-S16 represent the augmented samples from S01-S08, respectively.

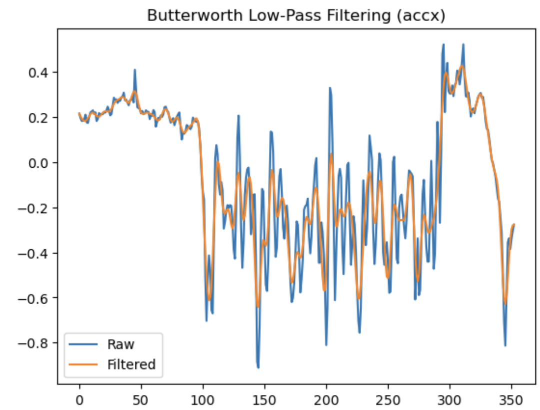

Fig. 1. Illustration of the effect of the Butterworth low-pass filter applied to the raw signal, showing that noise peaks are effectively attenuated while the primary motion pattern remains intact.

III. Model

A. Signal Preprocessing with Low-Pass Filter

To reduce high-frequency noise and obtain smoother acceleration signals, a fourth-order Butterworth low-pass filter was applied to all three accelerometer axes. The cutoff frequency was set to 3 Hz, with a sampling rate of 25 Hz. This configuration preserves the motion components relevant to human activities while suppressing unwanted vibrations and sensor noise. The filtered signals were then used for subsequent acceleration magnitude computation.

B. Threshold-Based Fall Detection Framework

This study employs a lightweight threshold-based model for fall detection. The model computes acceleration magnitude:

$$A = \sqrt{a_x^2 + a_y^2 + a_z^2} \tag{1}$$

Parameters such as impact threshold, motionless threshold, angle threshold, post-impact duration, and motionless duration are critical to the performance of the algorithm. Post-impact duration and motionless duration are fixed at 1000 ms, while the remaining three thresholds mentioned above were tuned through the Optuna-based optimization process. The algorithm monitors the acceleration magnitude A, and a fall is detected when the following conditions are sequentially satisfied:

- A exceeds the impact threshold.
- After 1000 ms (post-impact duration), the signal still remains below the motionless threshold for 1000 ms (motionless duration).
- The orientation angle between pre-impact and post-impact acceleration exceeds the angle threshold.

C. Leave-One-Out Cross-Validation

In this research, leave-one-out cross-validation was used in the optimization process. In each fold, one subject was held out as the test subject, and all remaining subjects were used for training and threshold optimization. This process was repeated for every subject in the dataset. This method prevents overfitting and captures the variability among subjects.

TABLE I
PER-SUBJECT OPTIMAL THRESHOLD SET (MEAN = GLOBAL THRESHOLD SET)

Subject	Impact Thresh (g)	Motionless Thresh (g)	Angle Thresh (deg)
S01	1.914754	0.452998	7.0000
S02	1.925098	0.454651	7.0000
S03	1.921422	0.457702	6.0000
S04	1.895973	0.457435	7.0000
S05	1.935543	0.455990	8.0000
S06	1.942502	0.448721	7.0000
S07	1.921057	0.451901	7.0000
S08	1.947043	0.459671	8.0000
S09	1.897989	0.453269	7.0000
S10	1.919284	0.452759	7.0000
S11	1.942431	0.461188	5.0000
S12	1.940026	0.453404	13.0000
S13	1.939477	0.444862	13.0000
S14	1.942180	0.452998	7.0000
S15	1.940165	0.462044	7.0000
S16	1.899455	0.237011	7.0000
Mean	1.926525	0.440846	7.6875

TABLE II
OVERALL CLASSIFICATION REPORT (GLOBAL THRESHOLD)

Class	Precision	Recall	F1-score	Support
ADL	0.84	0.78	0.81	566
Fall	0.74	0.81	0.78	442
Accuracy			0.79	1008
Macro Avg	0.79	0.80	0.79	1008
Weighted Avg	0.80	0.79	0.80	1008

TABLE III
FALL-TYPE DETECTION PERFORMANCE (GLOBAL THRESHOLD)

Fall Type	TP	FN	Support	Sensitivity (%)
F01 (Sitting → Fall Left)	66	12	78	84.62
F02 (Sitting → Fall Backward)	43	27	70	**61.43**
F03 (Standing → Fall Backward)	66	14	80	82.50
F04 (Standing → Fall Forward)	73	3	76	96.05
F05 (Sitting → Fall Right)	58	20	78	74.36
F06 (Tripped → Fall Forward)	53	7	60	88.33
Total / Mean	359	83	442	81.22

D. Hyperparameter Optimization via Optuna

The threshold parameters are tuned automatically using the Optuna optimization framework within each fold of the leave-one-out cross-validation (LOOCV). It performs multiple trials to minimize classification error by tuning impact, motionless, and angle thresholds. Each fold's optimal parameters were recorded, and after completing the optimization for all folds, the average values were computed to form a global threshold set that generalizes across subjects. This global threshold set was used to perform the final evaluation on the entire dataset.

E. Implementation and Pipeline

The entire pipeline is implemented in Python with two main scripts. The first one implements the main detection algorithm, including acceleration magnitude computation and threshold comparison. The second one is a Jupyter notebook which integrates data processing, parameters optimization, global threshold set computation, generation of classification outputs for both fall and ADL events.

IV. SUMMARY OF RESULTS

A. Overall Performance (Global Threshold Set Evaluation)

The proposed threshold-based model was evaluated using a global threshold set derived from the mean of threshold sets from all subjects. This configuration provides a generalized threshold set that can be applied to unseen data without re-training. Table I summarizes the tuned threshold sets from each subject and their mean values. The global impact, motionless, and angle threshold are 1.926525 (g), 0.440846 (g), 7.6875 (deg), respectively. Table II presents the classification report obtained by applying the global threshold set to the entire dataset, achieving an overall accuracy of around 79%.

B. Activity-Level Analysis

To analyze the causes of misclassification, two tables for both fall types and ADL types were generated. Table III shows the detection performance for each type of fall. Most fall types achieved over 80% sensitivity, except "F02 (Sitting and Fall Backward)", which shows that Falls without distinct acceleration spikes were more difficult to detect. These results indicate that, while the model is capable of detecting high-impact activities, variability in fall dynamics reduces its ability to generalize evenly across all fall scenarios. Table IV shows the detection performance for each type of ADL. Many misclassifications occurred mainly in those ADLs that include large arm movements such as "A05 (Jumping and Reaching)" and "A07 (Jogging)". In contrast, ADLs with relatively smaller arm movements, such as "A01 (Sitting and Standing)" and "A02 (Walking)", were almost perfectly detected.

This finding confirms that ADLs with larger arm motions have acceleration peaks similar to falls, demonstrating that fixed thresholds are less reliable for wrist-worn sensors.

V. CONCLUSION AND FUTURE WORK

This study evaluated a lightweight threshold-based fall-detection algorithm using wrist-worn accelerometer data collected from the WeBe Band. A global threshold set was derived from the mean of the threshold sets from all leave-one-out cross-validation (LOOCV) folds after the optimization process. This configuration enables subject-independent fall detection while maintaining computational efficiency suitable for embedded deployment. The results demonstrated that this model achieved an accuracy of around 79%.

Previous research has shown that sensor placement affects fall detection performance [3]. Sensors placed at trunk produce more stable signals, while sensors placed at wrist may be disrupted by large arm movements that can also have high acceleration spikes, exceeding the thresholds. Our results align with this perspective, as activities involving intense arm movements (e.g., jogging, jumping and reaching) were frequently misclassified as falls. Thus, wrist-based placement poses inherent challenges for threshold-based detection due to greater motion variability.

Another limitation involves the use of simulated falls performed by young participants on an inflatable mattress, which differ from real-world fall conditions among elderly [5], [7]. Real falls experienced by elderly are typically slower and

TABLE IV
ADL-TYPE DETECTION PERFORMANCE (GLOBAL THRESHOLD)

Activity	TN	FP	Support	Specificity (%)
A01 (Sit ↔ Stand)	79	1	80	98.75
A02 (Walking)	77	3	80	96.25
A03 (Walking down stairs)	61	9	70	87.14
A04 (Walking up stairs)	69	1	70	98.57
A05 (Jumping and Reaching)	19	51	70	**27.14**
A06 (Clapping)	63	3	66	95.45
A07 (Jogging)	18	52	70	**25.71**
A08 (Tie shoe laces crouch)	56	4	60	93.33
Total / Mean	442	124	566	78.09

involve protective reactions, producing smaller acceleration peaks. Moreover, inflatable mattresses absorb the impact, reducing acceleration peaks. These factors suggest that the current dataset, while useful for controlled evaluation, may overestimate the algorithm's sensitivity compared to real-world deployments.

To ensure higher performance of this model with the same settings, future work should focus on improving generalization and real-world validation by expanding data collection to include older adults and natural fall events under realistic conditions.

Overall, this study shows both the potential and limitations of using a simple threshold-based model for fall detection on wrist-worn device. While the model is computationally efficient, enhancing cross-subject robustness and reliability in real-world wrist-worn settings remains a critical direction for future research.

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